

SCI-MAP

Ordinary Mapmaking and Observation Coaddition

Formal Scientific/Engineering Contract

Scientific contract version: **v0.1**
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Authority boundary. This is the normative formal view of the implementation-independent scientific contract. It preserves the canonical equations, 52 requirements, 25 predictions, exact state and support semantics, provenance, and conformance routing. The separate science-team rationale r0.5 explains the same science without replacing this authority. This document does not claim implementation conformity, representation fidelity, scientific validation, observational performance, or production readiness.

Physical model and contract at a glance

MAP consumes the realized MAP-facing output of PTC; it never bypasses PTC to consume CAL output directly. ALIGN/AST supplies the coordinate facts and MAP supplies the target grid and declared projection. SCI-MAP applies one fixed, positive-coefficient linear averaging operator to the signal and to every declared linear companion. A coadd then applies a second positive-coefficient operator to atomically admitted, compatible observation bundles on the same grid. All scientific statements are conditional on the complete admitted calibration, alignment, eligibility, coefficient, response, geometry, and covariance state.

Facet	Compact authority
Input	Realized SCI-PTC v0.1/r0.5 calibrated- x -derived nonpolarimetric total-intensity-equivalent product in mJy/beam , with exact originating nominal-beam identity, occurrence/generation, retention, causes, and response state; exact AST RTC-grid coordinates for the same stable sample and parent; and a future owner-selected PTC MAP coefficient.
Output	A complete immutable observation-map bundle and, when requested and admissible, a complete centered-integer common-grid observation coadd bundle.
Equation	On admitted membership, $N_p = \sum_i a_{pi} z_i$ and $Q_p = \sum_i a_{pi}$. The normalized scientific vector contains exactly the rows authorized by the effective support policy, with $\hat{m}_p = N_p/Q_p$ there; response and covariance use the same explicit row selection. Coadd repeats this restricted structure over complete observations.
Source	Owner-approved Scope Brief and prior frozen core, amended by the owner-authorized PTC-to-MAP handoff correction recorded in CHANGE_LOG_R0.4.md .
Status	Contract author draft. Algebraic predictions are preregistered below; none has been executed here. All stronger claim layers remain unassessed.

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1 Scientific boundary and estimand

The ordinary observation map estimates the coefficient-weighted average of the admitted realized PTC sample quantity in each declared map pixel. Its meaning is inherited through the required CAL-to-PTC route, from ALIGN/AST's coordinate relation, and from PTC's transformed signal, product identity, availability, producer-local validity, and MAP-facing coefficient. SCI-MAP does not repair or reinterpret those facts. VAL may register and evaluate a named rule, but it is not the author of producer or MAP policy.

Let i bind observation, detector occurrence and UID, stable RTC output sample n , exact PTC product/application generation, segment, and array/network/group; time is an attribute and storage positions are locators. Here $z_i = Z_i^{\text{PTC}}$ is the realized PTC transformed signal and γ_i is its exact PTC-owner-selected MAP-facing coefficient. The paired x/r symbols remain KID readout coordinates. The effective coefficient is $a_{pi} = G_{pi}\gamma_i$. No such γ family is selected in frozen PTC r0.5, so the ordinary numerical route is unavailable rather than defaulting to unity, loading, sensitivity, scatter, or precision.

The fixed contribution set $\mathcal{C}_p$ factors transformed-signal availability, PTC retention, coefficient availability, coefficient/QC, SCI-MAP:map_upstream_admission@1, AST validity, finite signal, finite positive coefficient, admitted projection/boundary, and MAP-local companion gates. Each gate is screened before payload evaluation and preserves its knowledge state, cause, and failure scope. Let Π_{eff} be the recorded effective product support policy and $\mathcal{S}_{\text{out}}(\Pi_{\text{eff}})$ the exact set of map rows for which every support predicate that policy designates applicable passes. This is a row domain, not a numerical mask. The frozen ordinary result is then reused directly:

$$a_{pi} = G_{pi} \gamma_i, \tag{1}$$

$$\mathcal{C}_p = \{i : S_Z(i) \wedge R_{\text{PTC}}(i) \wedge A_\gamma(i) \wedge Q_\gamma(i) \wedge A_{\text{MAP}}(i) \wedge V_{\text{AST}}(i) \\ \wedge F_Z(i) \wedge P_\gamma(i) \wedge P_G(i, p) \wedge L_{\text{MAP}}(i, p)\}, \tag{2}$$

$$\mathcal{S}_{\text{out}}(\Pi_{\text{eff}}) = \{p : \text{authorize}_{\Pi_{\text{eff}}}(p) = 1\}, \tag{3}$$

$$N_p = \sum_{i \in \mathcal{C}_p} a_{pi} z_i, \quad Q_p = \sum_{i \in \mathcal{C}_p} a_{pi}, \tag{4}$$

$$K_p = \sum_{i \in \mathcal{C}_p} a_{pi} k_i, \tag{5}$$

$$\hat{m}_p = N_p / Q_p, \quad \hat{k}_p = K_p / Q_p, \quad p \in \mathcal{S}_{\text{out}}, \tag{6}$$

The supported division is not a recipe for filling unsupported pixels. Raw accumulators and representation-specific storage slots may exist outside $\mathcal{S}_{\text{out}}$, but the normalized scientific vector has no row there. A dense zero, sentinel, or coincidentally finite payload is not a semantic substitute and is not measured zero sky. Likewise, Q_p is a normalization fact. Its dimensions or historical name do not make it precision.

The active role is a calibrated- x -derived nonpolarimetric total-intensity-equivalent quantity in **mJy/beam**. The exact originating fixed-nominal-beam/template and CAL/PTC lineage are part of the quantity identity. Neither the unit nor a STOKES token promotes it to Stokes I. V0.1 does not authorize polarimetry, measured-R mapmaking, JINC, maximum-likelihood mapmaking, mosaicking, or reprojection.

2 Linear operator, response, and absolute mode

For selected transformed samples $\mathbf{z}$, projection matrix G , diagonal coefficient matrix Ω , forward sky-to-sample operator H , and exact effective output-row selector J_{out} , the map operator and its conditional response and covariance are:

$$D_Q = \text{diag}(G\Omega\mathbf{1}), \quad J_{\text{out}}\mathbf{z} = (z_p)_{p \in \mathcal{S}_{\text{out}}}, \quad D_{Q,\text{out}} = J_{\text{out}}D_QJ_{\text{out}}^\top, \quad (7)$$

$$A_{\text{out}} = D_{Q,\text{out}}^{-1}J_{\text{out}}G\Omega, \quad \widehat{\mathbf{m}}_{\text{out}} = A_{\text{out}}\mathbf{z}, \quad (8)$$

$$R_{\text{out}} = A_{\text{out}}H, \quad \delta\mathbb{E}[\widehat{\mathbf{m}}_{\text{out}} | \Theta] = R_{\text{out}}\delta\mathbf{s}, \quad C_{m,\text{out}} = A_{\text{out}}N_{\text{PTC}}A_{\text{out}}^\top. \quad (9)$$

The diagonal $D_{Q,\text{out}}$ contains only finite positive normalizations on $\mathcal{S}_{\text{out}}$, so its ordinary inverse is defined. The matrix A_{out} has no rows outside that set. Extending it to the full storage grid by writing zeros would be a representation convenience, not this scientific operator, response, or covariance.

Here N_{PTC} is a declared covariance of PTC-domain processed-sample error when such information is available, not the ensemble covariance of sky plus error. The state Θ contains all selected or estimated calibration, alignment, membership, coefficients, response, geometry, identity, and effective-support facts. Equation (9) therefore makes a conditional perturbation statement on exactly the authorized rows; it makes no unnamed μ_0 or residual-bias claim. If any element of Θ is estimated from the same data, an unconditional claim requires its joint uncertainty and cross-covariances.

The forward object H is typed as a complete fixed-state upstream-to-PTC response, PTC full-procedure response, whole-chain injection, or another exact named family. A realized PTC-grid companion begins at MAP and is not passed through upstream response again. When the required forward information is available, the response matrix $R_{\text{out}} = A_{\text{out}}H$ describes what sky structure survives on the authorized output domain for its declared model, domain, and limitations. A stored kernel image is the realized response to a named unit-bearing template $\boldsymbol{\tau}$ only if its sample-domain parent is $\mathbf{k} = H\boldsymbol{\tau}$ and the stored map is $A_{\text{out}}\mathbf{k} = R_{\text{out}}\boldsymbol{\tau}$ with the same exact row identity as signal. Without that identity it is a tracer, not the measured response. If any signal member lacks the required exact parent, response is limited or unavailable on the unchanged signal membership; no hidden subset, zero, or identity array can stand in for it. That state does not invalidate the numerical map or prohibit later analysis. A later simulated response or response-corrected map is a new versioned product bound to the original MAP product and processing identity; it does not rewrite the original product's claim.

V0.1 fixes the signal-centering operator to $L = I$. SCI-MAP therefore does not remove a mean or a constant mode during coaddition. It preserves a constant admitted sample vector on the exact support-authorized output rows because numerator and denominator use identical coefficients. A stronger constant-sky statement additionally requires the forward operator to map that sky mode to the admitted sample constant. Neither result alone establishes point-source, integrated-flux, extended-source, edge, or absolute astrometric fidelity. Any The PTC nonrestored additive reference λ , fitted correlated removed component $\widehat{U}_{\text{corr}}$, total removed component $U_{\text{total},\Theta}$, removed-subspace identity, fixed-state null space, full-procedure invariant/unidentifiable modes, and any separately authorized residual conditional bias are distinct facts. Any absolute mode already removed upstream remains absent and must be carried in response and null-mode provenance.

3 Input classes and falsifiable response behavior

The same response equation yields distinct, testable input classes without granting them equal scientific meaning:

Constant.

Fixed positive coefficients preserve a constant admitted sample vector even when coverage and variance vary.

Delta or basis pixel.

The output is a support-row-restricted column of R_{out} , including beam, projection, preprocessing, and boundary effects.

Point-source template.

The output is $R_{\text{out}}\tau_b$ on exactly the authorized rows. Peak, integrated, and fit amplitudes are different functionals.

Resolved, gradient, and Fourier structure.

The output is $R_{\text{out}}\tau$; constant preservation says nothing about these modes.

Variable coverage.

Fixed-coefficient normalization may preserve a constant while covariance and nonconstant response vary spatially.

Finite boundary.

Truncation can preserve a local constant yet change a compact-source peak or integral. No wrap or clamp is authorized.

The authorized base family is `SCI-MAP:one_hot_containing_pixel@1`. It uses half-open pixel cells, assigns unity to the unique finite in-grid containing pixel, excludes the outer upper boundary, never wraps or clamps, and has $\sum_p G_{pi} = 1$ only on that in-grid domain. Fractional projection remains a future role; no kernel, conservation, boundary-loss, response, or covariance semantics are inferred for it.

Response fidelity is therefore a validation layer, not a consequence of code executing the algebra. Delta, point-template, extended, Fourier, variable- coverage, and edge injections are preregistered in Appendix D; they are not run or claimed here.

4 Conditional uncertainty and statistical labels

For independent selected sample errors with a future exact coefficient family proven to be true inverse variance, the general formal relation is:

$$\text{Var}(\hat{m}_p \mid \Theta) = \frac{\sum_i G_{pi}^2 \gamma_i}{(\sum_i G_{pi} \gamma_i)^2}, \quad \text{Cov}(\hat{m}_p, \hat{m}_r \mid \Theta) = \frac{\sum_i G_{pi} G_{ri} \gamma_i}{Q_p Q_r}, \quad p, r \in \mathcal{S}_{\text{out}}, \quad (10)$$

$$\phi_p = \frac{Q_p^2}{\sum_i G_{pi}^2 \gamma_i}, \quad \phi_p = Q_p \iff \sum_i \gamma_i (G_{pi}^2 - G_{pi}) = 0, \quad p \in \mathcal{S}_{\text{out}}, \quad (11)$$

Equation (11) states why normalization is not precision by default. One-hot geometry satisfies the projection part of the equality, but cannot establish that γ is calibrated inverse variance or that sample errors are independent. Frozen PTC r0.5 has selected no γ family.

Four uncertainty objects must not be conflated:

1. Q , an ordinary normalization with its declared gridding unit;
2. $\phi = 1/[C_{x,\text{out}}]_{pp}$, a conditional marginal inverse variance under its named assumptions;
3. $C_{x,\text{out}}$ or $C_{x,\text{out}}^+$, the full conditional covariance or precision on the identifiable stochastic subspace of the exact authorized row domain; and
4. an empirical covariance, weight, or significance calibrated by NOI.

The bundle reports what uncertainty or covariance information it actually provides, its meaning, domain, assumptions, and limitations, and whether fuller information is persisted, lineage-resolvable, summarized, incomplete, or unavailable. Missing correlations are named, not set to zero. Neither normalization, hits, exposure, nor diagnostic weight is silently promoted to covariance. Absence of a complete covariance model does not invalidate the map or prohibit later scientific analysis. A later covariance estimate is a new versioned product bound to the original map and processing identity. The minimum persistence form remains an owner decision.

5 Distinct map facts, support, and MAP-local validity

The base/unfiltered MAP bundle separates facts because they answer different questions:

Fact	Meaning
AST-geometric incidence	Which coordinate-valid occurrences touch the unique containing pixel, independent of signal availability.
MAP-route-candidate incidence	Which exact realized PTC occurrences also pass the occurrence-level MAP upstream-admission profile.
Estimator-admitted contributions	How many occurrences passed every ordinary coefficient and contribution predicate.
Admitted observation count	How many complete observation bundles contribute to a coadd pixel; not sample hits or attempted observations.
Upstream-eligible exposure	Named exposure over the source-available, coordinate-valid projected population, before MAP estimator-specific rejection.
Retained exposure	The same named exposure over estimator-admitted membership and placement.
Normalization support	The numerical threshold predicate used to authorize ordinary division.
Science-policy support	The separately stricter adopted policy predicate.
MAP-local base-product validity	MAP's conjunction of admitted identity, support, finite signal, and no MAP-product failure; not downstream eligibility.

No one fact is an alias for another. In particular, exposure is not precision, positive normalization is not final validity, and a finite payload is not evidence of scientific availability.

ALIGN owns e_{acq} and valid-original e_{vo} , in seconds of physical native integration support on stable original occurrences. RTC, CAL, and PTC preserve those facts and their causes. MAP defines upstream-eligible, retained, projected, and coadded exposure by unioning and deduplicating exact original-occurrence parents, then applying the one-hot placement and atomic coadd admission. Invalid-original has $e_{\text{vo}} = 0$; synthesis, replacement, overlapping support, cadence, hits, Q , and γ create no exposure. An unavailable lineage makes the affected exposure role unavailable rather than reconstructible from duration or count.

The adopted v0.1 threshold rule is policy authority, not a derived physical law. For the declared plane and policy population of finite positive Q_p values, sorted in ascending order, the admitted formula names $c = \text{coverage_cut}$:

$$\mathcal{I}_{\Pi_{\text{eff}}} = \text{the declared support-policy population,} \quad (12)$$

$$\mathcal{P} = \{Q_p : \text{finite}(Q_p), Q_p > 0, p \in \mathcal{I}_{\Pi_{\text{eff}}}\}, \quad N = |\mathcal{P}|, \quad (13)$$

$$k = \left\lfloor \frac{\lfloor 0.75N \rfloor + N}{2} \right\rfloor, \quad Q_{\star} = \begin{cases} \mathcal{P}_{(k)}, & N > 0, \\ 0, & N = 0, \end{cases} \quad (14)$$

$$c = \text{coverage_cut (dimensionless scalar)}, \quad \text{admit}_{\Pi_{\text{eff}}}(c) = 1, \quad (15)$$

$$T_{\text{norm}} = Q_{\star}c/10, \quad T_{\text{sci}} = Q_{\star}c, \quad (16)$$

$$S_p^{\text{norm}} = \mathbf{1}[\text{finite}(Q_p), Q_p > 0, Q_p \geq T_{\text{norm}}], \quad (17)$$

$$S_p^{\text{sci}} = \mathbf{1}[\text{finite}(Q_p), Q_p > 0, Q_p \geq T_{\text{sci}}]. \quad (18)$$

Both predicates require the pixel's own explicit finite positive coefficient; the empty-population threshold of zero does not manufacture support. The operator row set $\mathcal{S}_{\text{out}}$ is the exact intersection of the predicate row sets designated applicable by Π_{eff} ; it is not a full-grid mask with zero semantics.

The scientific owner has fixed $c = \text{coverage_cut}$ as a dimensionless support-policy scalar because Q_p , $Q_{\star}$, and both thresholds are directly compared in the same coefficient unit. SCI-MAP-CI-001 records the resolved inconsistency and the bounded amendment to the canonical equations, REQ-031/032, and PRED-012. This disposition does not establish a numerical domain. Negative, zero, non-finite, and greater-than-one values, the recommended range, policy authority, and failure behavior remain SCI-MAP-OD-007. Threshold rationale and future change authority remain SCI-MAP-OD-001 and SCI-MAP-OD-002.

6 Observation coaddition on one common grid

Coaddition consumes immutable complete observation bundles, not independent planes. `SCI-MAP:observation_coadd_admission@1` atomically checks quantity and nominal-beam convention, PTC route/product role, response source/class, complete AST frame/WCS/grid, support policy, coefficient family, covariance disclosure, null/additive-reference state, required companions, lifecycle, and parentage before any coadd-owned state changes.

The only v0.1 placement is centered integer embedding on the same grid. Observation and coadd row and column differences are nonnegative and even; their half-differences are the integer offsets, and the shifted reference pixels identify the same world coordinate. The lossless typed WCS is the authority. No fractional shift, interpolation, reprojection, wrapping, or implicit source recentering is allowed.

For the atomically admitted per-pixel set $\mathcal{O}_p$, every contributing observation row is already a member of that observation's support-authorized scientific domain. Let $\mathcal{S}_{\text{out}}^c$ be the exact rows authorized by the effective coadd support policy. The MAP-owned family `SCI-MAP:uniform_observation_coadd_coefficient@1` assigns dimensionless $u_{op} = 1$ to every admitted row. It is equal-observation weighting, not precision. The coadd is:

$$\mathcal{S}_{\text{out}}^c = \{p : \text{authorize}_{\Pi_{\text{eff}}^c}(p) = 1\}, \quad (19)$$

$$u_{op} = 1 \quad \text{for each admitted observation row}, \quad (20)$$

$$N_p^c = \sum_{o \in \mathcal{O}_p} u_{op} \hat{m}_{op}, \quad Q_p^c = \sum_{o \in \mathcal{O}_p} u_{op}, \quad (21)$$

$$\hat{s}_p = N_p^c / Q_p^c, \quad p \in \mathcal{S}_{\text{out}}^c, \quad (22)$$

$$J_{\text{out}}^c \mathbf{z} = (z_p)_{p \in \mathcal{S}_{\text{out}}^c}, \quad (23)$$

$$\hat{\mathbf{m}}_{\text{out}}^c = (\hat{s}_p)_{p \in \mathcal{S}_{\text{out}}^c} = B_{\text{out}} \hat{\mathbf{m}}_{\text{obs}}, \quad (24)$$

$$\hat{k}_p^c = \frac{\sum_{o \in \mathcal{O}_p} u_{op} \hat{k}_{op}}{Q_p^c}, \quad p \in \mathcal{S}_{\text{out}}^c. \quad (25)$$

$$C_{c,\text{out}} = B_{\text{out}} C_{\text{obs}} B_{\text{out}}^T, \quad R_{c,\text{out}} = B_{\text{out}} T_{\text{obs}}. \quad (26)$$

Here $\hat{\mathbf{m}}_{\text{obs}}$ is the stacked signal on exact observation-row identities; B_{out} is the centered-integer equal- observation averaging operator; T_{obs} stacks exact compatible observation response objects; and C_{obs} is the block covariance with every available within- and cross-observation block explicitly located. Unknown blocks are not zero. The coadd operator B_{out} has rows only in $\mathcal{S}_{\text{out}}^c$. Coadd accumulators or dense storage slots outside that set do not become a zero-valued scientific coadd. On its exact domain the coadd is a convex weighted mean. One admitted observation reproduces itself; two admitted observations produce their equal-observation mean. Its normalization is not precision. With cross-observation covariance, the exact conditional covariance retains those blocks on the same selected rows. A correlated-GLS benchmark may require negative combination coefficients, placing it outside this ordinary positive-coefficient contract.

Available response, retained exposure, and admitted observation count follow the same bundle admission and placement. If any member response is unavailable, the response-independent base-coadd role may retain the numerical signal but must report coadd response unavailable; it forms no hidden response subset. A rejected observation changes no coadd state anywhere, even if a subset of its planes would otherwise appear compatible.

7 Identity, WCS, units, and lifecycle

A map cannot be reconstructed from its collection position. Paired PTC x/r components remain upstream detector coordinates and are carried only when a MAP-facing fact requires their identity; MAP does not recast r as a response. The minimum identity includes the nonpolarimetric total-intensity-equivalent product role; external observation or coadd ID; array/group and occurrence domains; estimator, projection, PTC and coadd coefficient families; response/covariance states; signal unit and exact nominal-beam parent; complete AST frame/WCS; exposure convention; MAP profiles; exact PTC/AST generations and product identity. A STOKES field is only a locator unless separate quantity authority exists. The policy must be scientifically identifiable, durable, and comparable, but this contract prescribes no JSON, hash, or sidecar serialization. Array ID, name, index, network, map group, map index, detector occurrence, and container position remain distinct.

Science maps bind the complete frame identity supplied by `SCI-AST:rtc_output_grid_coordinates@1`, including the actual reference system, equinox/epoch, time scale, site/model state, TAN topology, WCS, axis order, signs, handedness, and degree-valued axes. MAP never equates J2000, FK5, ICRS, apparent, observed, or unrefracted frames by name/default. A Pointing/OOF ordinary-map use, if the owner registers it, uses a declared AltAz tangent plane in arcseconds; its mode-specific meaning is outside SCI-MAP. FITS/WCS coordinates are one-based and in-memory row/column indices are zero-based.

Full-precision typed or sidecar WCS is the lossless admission and provenance authority. A physical FITS serialization may differ by at most 0.1 arcsec in maximum sky separation while preserving exact axis sign, handedness, orientation, and centered-integer shape/reference-pixel relations. This is a representation tolerance, not authority for a different grid.

Requested state records the immutable accepted intent. Effective state records the executable supported plan. Observation-resolved state replaces all observation-specific arrays, mappings, geometry, coefficients, and provenance. Realized state records what actually ran, all accumulators and distinct facts, products, offsets, representation states, cardinality, and failures. State flows one way; execution does not rewrite the request or inherit a prior observation's resolution.

8 Failure, downstream boundary, and immutable parentage

Failure semantics protect the scientific bundle from partial truth. An invalid candidate can be excluded at contribution scope; an unsupported pixel is raw-invalid; an incompatible observation is rejected atomically; an unrepresentable finite aggregate, coordinate/index conversion, missing required companion, or failed required publication propagates at its declared product scope before live state mutation. A completion marker cannot override a missing required output.

No downstream package receives a bare signal plane as a complete map. The boundary includes the estimator and normalization identity; response state; conditional covariance/formal-weight state and assumptions; distinct map facts; additive-bias and null-mode state; units; WCS; grouping; lifecycle; raw parentage; dependency versions; and failures. NOI may propagate a fixed operator but owns realization construction and empirical calibration. FLT owns filtered support, response, covariance, and validity. SRC, MODE, and BEAM own inference and fit uncertainty. FRUIT owns recurrence and iteration state.

Every later analysis is responsible for deciding which facts its own claim requires. An unavailable response prevents the original MAP bundle from making an unsupported response-dependent claim, but it does not prohibit analysis or invalidate other map claims. A significance analysis cannot relabel merely formal standardization; an absolute-level claim cannot ignore an unobservable offset. This contract defines no closed-world registry of all future uses. A downstream product may have local validity or eligibility, but it cannot rewrite MAP-local base-product validity or sever the immutable base/unfiltered MAP-parent identity.

When an admitted noise realization is described as using the same operator, membership, projection, grouping, coefficients, normalization, exact support-authorized row selection, boundary, and coadd admission are fixed or identically prescribed. Recomputing them from randomized data creates a different estimator, regardless of matching array shapes or filenames.

9 Uncertainty, validity, validation, and claim layers

The contract deliberately separates four reasoning layers:

1. **Uncertainty layer:** conditional covariance and response are propagated through the fixed operators; omitted state uncertainty remains explicit.
2. **Validity layer:** contribution eligibility, numerical support, science-policy support, and MAP-local base-product validity determine the exact rows on which the estimator is scientifically available; unsupported rows are absent, not zero-filled observations.
3. **Validation layer:** preregistered analytic fixtures, matrix comparisons, injections, covariance controls, representation round trips, and observational references test distinct claims.
4. **Readiness layer:** implementation evidence, operational reliability, data completeness, and release authority are assessed elsewhere.

Claim layer	Status here	Evidence needed before an achieved claim
Algebraic scientific contract	Proposed author draft	Scientific-owner review and contract freeze; analytic predictions are specified, not executed.
Engineering conformance	Not assessed	Requirement-by-requirement inspection and tests against one exact candidate identity.
Representation/response fidelity	Not assessed	WCS round trips and delta, template, extended, variable-coverage, and edge injections with preregistered bounds.
Observational performance	Not assessed	Independent astronomical amplitude, shape, position, covariance, and repeatability references where available.
Production readiness	Not assessed	Complete required artifacts, failure/log audit, operational gates, owner release decision, and no disqualifying gaps.

Floating reductions require a preregistered bound derived from operation count and conditioning. For a sum of n terms, a reference policy may use $\gamma_n = n\epsilon/(1 - n\epsilon)$ and a fixed multiple such as the admitted core's $8\gamma_n \sum_i |a_i|$, frozen before observing results. Integer counts, identities, masks, support, offsets, and product cardinality remain exact. Tolerances are not relaxed after failure.

The contract's 25 predictions are listed formally in Appendix D. No validation run, implementation inspection, observational reduction, or production decision was performed during this scientific authorship stage.

10 Open owner decisions and scientific disposition

Nine decisions remain intentionally unresolved: (1) the physical rationale for the adopted support thresholds; (2) future threshold change authority and evidence; (3) the minimum response information that the original MAP product must provide for each response-dependent claim it makes; (4) the minimum covariance representation that must persist rather than remain lineage-resolvable; (5) whether physical observation-map publication is an abstract obligation beyond the effective v0.1 required-product rule; (6) whether Pointing/OOF ordinary arithmetic is a registered reuse of this operator; and (7) the admissible numeric domain, boundary-case dispositions, and failure behavior for the dimensionless $c = \text{coverage_cut}$; (8) authorized MAP projection classes, normalization, and boundaries; and (9) canonical-grid preparation and future reprojection/mosaicking ownership. Appendix [A](#) reproduces the exact questions and conservative interim behavior from the owner ledger. SCI-MAP-CI-001 is resolved and OD-007 is correspondingly narrowed without renumbering.

Pending those decisions, the scientific contract is coherent at its stated conditional boundary. It authorizes ordinary positive-coefficient normalized mapmaking and atomic centered-integer observation coaddition only. It does not establish implementation conformity, representation fidelity, observational performance, or production readiness.

A Owner-decision register

This compact register is generated from `SCIENTIFIC_OWNER_DECISION_LEDGER.md`. The mechanical contract checker requires exact decision IDs, open statuses, decision prompts, and conservative interim dispositions to match before either PDF is accepted. Rendering this register does not answer or close an item.

Decision/status	Exact decision requested	Conservative draft behavior pending decision
SCI-MAP-OD-001 OPEN	What physical rationale is authoritative for the adopted v0.1 normalization-support and science-policy-support threshold rule?	State the formula exactly, label it adopted policy, and make no physical optimality, completeness, noise, or response claim from it.
SCI-MAP-OD-002 OPEN	Who may change the support-threshold rule after v0.1, and what evidence is required for a successor?	Freeze the v0.1 formula and population identity; treat any altered multiplier, quantile/index, population, or predicate as a contract change.
SCI-MAP-OD-003 OPEN	What response information must the original v0.1 MAP product provide for each response-dependent claim it makes, and which original product claims remain unavailable when response information is incomplete or unavailable?	Report the actual response state, meaning, domain, and limits; fabricate neither zero nor identity response. Make no unsupported response-dependent claim. Permit later response or corrected-map products only with new versioned identity bound to the original MAP product and processing identity.
SCI-MAP-OD-004 OPEN	What minimum covariance representation must persist in a raw bundle, and what may remain resolvable through lineage?	Report what uncertainty/covariance information exists, with meaning, domain, assumptions, limitations, omissions, and lineage. Do not promote normalization, hits, exposure, or diagnostic weight. Permit later estimates as new versioned products bound to the original.
SCI-MAP-OD-005 OPEN	Beyond the adopted v0.1 effective required-output rule, is physical observation-map publication an abstract contract obligation whenever coaddition is requested?	Preserve the v0.1 rule that every product designated required by the effective plan, including required observation products during coaddition, must publish successfully. Do not generalize this into a future universal storage mandate.
SCI-MAP-OD-006 OPEN	Is Pointing/OOF ordinary-map arithmetic a registered use of the shared SCI-MAP v0.1 operator, with mode-specific interpretation remaining outside this package?	State only that reuse is conditional on a fully conforming input bundle; grant no mode-specific fitting, astrometric, response, or production authority.
SCI-MAP-OD-007 OPEN	What numerical domain is authoritative for the dimensionless scalar <code>c=coverage_cut</code> , including the required disposition of negative, zero, non-finite, and greater-than-one values, and what failure behavior applies when a numerical state is not authorized?	Assert no general numerical domain. Require the exact numerical value to be explicitly admitted by the owner-authorized effective support policy; otherwise fail closed before constructing support-authorized output rows or mutating a required product.
SCI-MAP-OD-008 RESOLVED	Which sample-to-pixel projection classes are authorized for ordinary SCI-MAP v0.1, what normalization applies to MAP-owned <code>G_pi</code> , and how is boundary loss represented?	Authorize only <code>SCI-MAP:one_hot_containing_pixel@1</code> with half-open cells, unity on the unique finite in-grid containing pixel, outer-upper-boundary loss, no wrap/clamp, and fractional projection deferred.
SCI-MAP-OD-009		

Decision/status	Exact decision requested	Conservative draft behavior pending decision
OPEN	Is upstream crop or pad to a canonical common grid an authorized preparation for v0.1 observation coaddition, which package owns it, and which future package owns reprojection or mosaicking beyond centered-integer placement?	Reject every odd-difference or otherwise incompatible observation bundle. SCI-MAP performs no crop, pad, fractional shift, reprojection, interpolation, or mosaic; no other package is named as authorized until the owner decides.

B Requirement and crosswalk summary

The detailed 52-row requirement crosswalk and 25-row prediction crosswalk remain in `CROSSWALK.md`. This compact index makes the rationale's traceability visible without duplicating the normative clause text or the detailed row table. Appendix C renders every canonical requirement, and Appendix D renders every prediction, directly from the shared authority.

Authority range	Scientific surface	Principal rationale / open gate
REQ-001	Version and revision identity	Cover; owner freeze
REQ-002–010	Capability, input, identity, projection, coefficients, lifecycle, and admission	Sections 1, 7, 8; upstream producer facts
REQ-011–018	Ordinary accumulators, exact support-row publication, response, and absolute mode	Sections 1–3; OD-003
REQ-019–024	Row-selected covariance, formal weight, nuisance state, and statistical labels	Sections 2, 4; OD-004
REQ-025–035	Separate incidence/contribution/exposure/support facts, threshold policy, exact rows, and MAP-local validity	Section 5; OD-001, OD-002, OD-007
REQ-036–042	Complete bundles, atomic centered-integer coadd, row-selected coadd	Sections 6, 8; OD-003, OD-004, OD-007
REQ-043–048	response/covariance, and GLS boundary	
REQ-043–048	Product identity, WCS, provenance, failure, and publication	Sections 7–8; OD-005, OD-006
REQ-049–052	Immutable raw parent, consumer boundary, fixed realization operator, and claim separation	Sections 8–9; OD-003, OD-004, OD-006, OD-007
PRED-001–025	Analytic consequences and conformance/representation challenges	Sections 3, 9; Appendix D; not executed
OD-001–009	Exact unresolved scientific-owner questions	Section 10; Appendix A; all OPEN
SCI-MAP-CI-001	Resolved dimensionless classification and bounded amendment	Section 5; equations, REQ-031/032, and PRED-012 amended in place

C Canonical requirements

This appendix is rendered directly from the shared canonical LaTeX authority. Requirement identifiers are stable within SCI-MAP v0.1. Editing prose in this document does not silently change them.

SCI-MAP-REQ-001 – Versioned authority. The scientific contract version shall be v0.1. The rendered document revision shall be recorded separately as r0.5; changing layout or exposition alone shall not be represented as a new scientific contract version.

SCI-MAP-REQ-002 – Capability and realized PTC input. The ordinary input shall be the realized SCI-PTC v0.1/r0.5 transformed calibrated- x product $z_i = Z_i^{\text{PTC}}$, a nonpolarimetric total-intensity-equivalent detector-time quantity in the inherited top-of-atmosphere, point-source-equivalent mJy/beam convention with its exact originating fixed-nominal-beam identity. Neither that unit nor a STOKES token establishes Stokes I. MAP has no CAL-to-MAP fallback, no inferred no-op PTC product, and no product when the required positive-rank PTC product is absent or PTC is disabled.

SCI-MAP-REQ-003 – Exact occurrence identity. Every candidate occurrence i shall bind observation, detector occurrence and UID, stable RTC output sample n , exact PTC product/application generation, segment, and array/network/group as applicable. Sample time is an associated attribute. Input-column, row, and container positions are locators only and shall not establish identity or equality.

SCI-MAP-REQ-004 – Producer facts and MAP admission. SCI-MAP shall consume PTC-owned transformed-signal availability, output-retention disposition, coefficient family/value/availability and coefficient-QC disposition without reinterpretation. It shall apply the occurrence-level profile SCI-MAP:map_upstream_admission@1; VAL Registry binds and VAL Core evaluates that MAP-authored policy but does not author it. Every fact, decision, cause, and scope remains separately typed.

SCI-MAP-REQ-005 – Coordinate and projection authority. MAP shall consume SCI-AST:rtc_output_grid_coordinates@1 for the same stable RTC sample n and complete parent chain as z_i ; it shall never join by time, row, cardinality, or coordinate equality. MAP owns G_{pi} through SCI-MAP:one_hot_containing_pixel@1: the unique finite in-grid pixel whose cell contains the coordinate under half-open bounds contributes $G_{pi} = 1$, every other pixel has $G_{pi} = 0$, and an outer-upper-boundary or out-of-grid coordinate contributes nowhere. Fractional projection is deferred. AST may materialize the relation only when the artifact binds the AST parent and exact MAP plan.

SCI-MAP-REQ-006 – Ordinary coefficient status. The ordinary route requires one exact PTC-owner-selected MAP-facing analysis/gridding coefficient family satisfying SCI-PTC-REQ-052–055 and the PTC-to-MAP boundary. No such family is selected in r0.5, so ordinary numerical MAP production is unavailable until that owner decision is versioned and bound. MAP shall infer neither unity, loading, sensitivity, scatter, nor precision; later coefficient generations shall not alter earlier PTC products, and cross-generation use requires explicit compatibility.

SCI-MAP-REQ-007 – Exposure lineage and accounting. ALIGN-owned occurrence facts e_{acq} and e_{vo} , in seconds of physical native integration support, shall remain immutable through RTC/CAL/PTC. MAP shall derive distinct upstream-eligible, retained, one-hot-projected, and coadded exposure from deduplicated original-occurrence lineage under its exact profiles. Invalid-original has $e_{\text{vo}} = 0$; synthesized, surrogate, missing, or unoccupied support adds no acquired exposure; replacement and overlapping support create no new exposure. Boundary loss contributes nowhere. Exposure shall not be reconstructed from duration alone, cadence, sample count, hits, Q , γ , or formal precision.

SCI-MAP-REQ-008 – Response information state. MAP shall report the response information it actually provides, including meaning, domain, parentage, normalization, units, and limitations, or an explicit typed unavailable state. Missing or incomplete response shall not invalidate an otherwise valid map, fabricate an identity response, or prohibit later scientific analysis. A later simulated response or response-corrected map shall be a new versioned product bound to the original MAP product and processing identity.

SCI-MAP-REQ-009 – Immutable plan and lifecycle. The accepted request shall be immutable and shall resolve one way through requested, effective, observation-resolved, and realized states. Disabled expert values remain inactive; one observation's resolved state shall not leak into another.

SCI-MAP-REQ-010 – Factored contribution predicate. For each pixel, membership in $\mathcal{C}_p$ shall require, as separately typed gates: transformed-signal availability; PTC output retention; coefficient availability; PTC

coefficient/QC disposition; MAP upstream admission; AST coordinate validity; finite z_i ; finite positive γ_i ; admitted projection/boundary relation; and MAP-local contribution plus required-companion predicates. Unknown, conflict, ineligible, unavailable, invalid, and not-requested all prevent arithmetic where applicable but preserve distinct causes and scopes. Every gate shall be screened before numerical payload evaluation; no numeric-zero mask represents the predicate.

SCI-MAP-REQ-011 – Ordinary accumulators. On the fixed contribution set, SCI-MAP shall accumulate $N_p = \sum_i a_{pi} z_i$, $Q_p = \sum_i a_{pi}$, and, when available for the identical membership, $K_p = \sum_i a_{pi} k_i$, where $a_{pi} = G_{pi} \gamma_i > 0$.

SCI-MAP-REQ-012 – Normalized publication. The scientific normalized output vector shall contain exactly the rows $p \in \mathcal{S}_{\text{out}}$ authorized by the applicable effective support policy. On those rows finite $Q_p > 0$ gives $\hat{m}_p = N_p/Q_p$, and an available response companion gives $\hat{k}_p = K_p/Q_p$. No unsupported row, zero-filled value, or sentinel is a normalized scientific output or measured zero.

SCI-MAP-REQ-013 – Accumulator interpretation. The numerator is a pre-normalized accumulator and Q_p is a normalization fact by default. Neither shall be labeled a sky estimator, inverse variance, confidence, coverage, or significance without the separately required conditions.

SCI-MAP-REQ-014 – Constant preservation. With fixed admitted membership and positive coefficients, the ordinary estimator shall preserve a constant admitted sample input on every row in its exact support-authorized output set. This prediction shall not create rows outside that set or be promoted to compact-source, extended-source, integrated-flux, absolute-offset, or astrometric fidelity.

SCI-MAP-REQ-015 – Projection-specific estimand. The authorized base projection is exactly SCI-MAP: `one_hot_containing_pixel@1`; it preserves unity over the finite in-grid domain and has explicit half-open boundary loss. Fractional projection, diagonal weighted least squares, and conservation across a finite boundary are outside this branch until separately authorized.

SCI-MAP-REQ-016 – Same operator for linear companions. Signal, available response/kernel tracer, and declared linear realizations shall use the same admitted membership, geometric placement, and exact support-authorized output-row selection; retained exposure and coadd observation count shall use the same membership and placement wherever their definitions require them.

SCI-MAP-REQ-017 – Realized response claim. A stored kernel may be called the realized response to template τ only when its sample parent satisfies $\mathbf{k} = H\tau$ and the support-restricted map product satisfies $\hat{\mathbf{k}}_{\text{out}} = A_{\text{out}} H\tau = R_{\text{out}} \tau$ on exactly the authorized rows. Otherwise it shall be labeled only a tracer or unavailable.

SCI-MAP-REQ-018 – Absolute-offset mode. SCI-MAP v0.1 shall apply no signal-centering operation: $L = I$. Coaddition shall not subtract a mean, remove a constant mode, or recenter a source; any upstream loss of that mode shall remain explicit in response and provenance.

SCI-MAP-REQ-019 – Conditional covariance. When a declared PTC-domain covariance N_{PTC} is available for the exact fixed membership and state Θ , MAP shall propagate it as $C_{m,\text{out}} = A_{\text{out}} N_{\text{PTC}} A_{\text{out}}^T$, retaining consequential off-diagonal terms. Missing, limited, or unavailable terms remain typed and are never zero-filled; absence of a complete model does not invalidate the signal map.

SCI-MAP-REQ-020 – When normalization is precision. Normalization may be called marginal inverse variance only when the exact selected γ family is calibrated true inverse variance, selected errors are independent on the asserted domain, and $\sum_i \gamma_i (G_{pi}^2 - G_{pi}) = 0$. One-hot geometry alone does not prove the coefficient premise.

SCI-MAP-REQ-021 – Formal diagonal weight. When its assumptions hold, $\phi_p = Q_p^2 / \sum_i G_{pi}^2 \gamma_i$ is the formal marginal inverse variance. It remains distinct from Q_p , support-restricted precision $C_{m,\text{out}}^+$, exposure, hits, and empirical statistical weight.

SCI-MAP-REQ-022 – Covariance and nuisance disclosure. The bundle shall report what uncertainty or covariance information it actually provides, its meaning, domain, assumptions, and limitations, and whether fuller information is persisted, lineage-resolvable, summarized, incomplete, or unavailable. It shall list omitted correlations and nuisance terms; absence of a complete covariance model shall not invalidate the map or prohibit later analysis, and unavailable information shall not be set to zero. A later covariance estimate may be attached only as a new versioned product bound to the original MAP product and processing identity.

SCI-MAP-REQ-023 – Data-derived state. If membership, coefficients, response, or geometry are estimated or selected from the same data, the displayed linear expectation and covariance shall remain explicitly conditional; unconditional bias or covariance requires a named joint model or resampling result outside this fixed-state claim.

SCI-MAP-REQ-024 – No significance promotion. Signal multiplied by the square root of merely formal weight shall be labeled formal standardized signal, not `sig2noise` or empirical significance. Empirical calibration remains NOI-owned.

SCI-MAP-REQ-025 – Distinct map facts. Every base/unfiltered MAP bundle shall separately represent AST-geometric incidence, MAP-route-candidate incidence, estimator-admitted contribution, admitted-observation count, upstream-eligible exposure, retained exposure, numerical-normalization support, science-policy support, and MAP-local base-product scientific validity. No fact or downstream eligibility decision shall become authority for another.

SCI-MAP-REQ-026 – Incidence layers. AST-geometric incidence records valid coordinate placement independent of signal availability; MAP-route-candidate incidence additionally requires the exact realized PTC route and occurrence-level upstream admission; estimator contribution additionally requires all remaining gates in $\mathcal{C}_p$. The one-hot touch convention counts the unique containing pixel only.

SCI-MAP-REQ-027 – Estimator contributions. Estimator-admitted contributions shall count exactly $i \in \mathcal{C}_p$. A candidate ineligible under the exact MAP admission or contribution policy for its preserved cause, or with unavailable/conflicting required state, zero/negative/non-finite coefficient, non-finite signal, or out-of-grid projection, contributes nowhere; no producer flag implies a universal action.

SCI-MAP-REQ-028 – Admitted observation count. For coadds, admitted observation count shall count complete observation bundles contributing at the pixel and shall not be confused with sample hits or observation attempts.

SCI-MAP-REQ-029 – Upstream-eligible exposure. MAP upstream-eligible exposure shall union and deduplicate the exact original occurrences carrying positive e_{vo} that influence MAP-route candidates admitted by `SCI-MAP:map_upstream_admission@1`, then place their seconds by the same one-hot containing-pixel rule. It precedes coefficient and MAP-local numerical rejection.

SCI-MAP-REQ-030 – Retained exposure. MAP retained exposure shall union and deduplicate positive- e_{vo} original occurrences influencing estimator-admitted members and place them by the same one-hot rule. Coadded exposure shall union compatible observation-level original-occurrence sets after atomic coadd admission; repeated parents are not double counted. Neither quantity is precision.

SCI-MAP-REQ-031 – Adopted threshold selector. For each declared support-policy population, sort its finite strictly positive normalization coefficients Q_p ; for count $N > 0$, select zero-based $k = \lfloor ([0.75N] + N)/2 \rfloor$ and $Q_\star = Q_{(k)}$; for empty input set $Q_\star = 0$. The exact population identity and exact dimensionless support-policy scalar $c = \text{coverage_cut}$ shall be recorded. The effective support policy shall explicitly admit that exact numerical c state before support resolution; otherwise the required product shall fail closed.

SCI-MAP-REQ-032 – Separate support predicates. Only after the effective support policy explicitly admits the exact value of the dimensionless scalar c , numerical normalization support shall require an explicit finite positive $Q_p \geq Q_\star c/10$, and science-policy support shall independently require an explicit finite positive $Q_p \geq Q_\star c$. This contract asserts no general numeric domain for c . Every non-authorized numerical state fails closed before a support row set is constructed. Neither support state shall establish final validity or precision.

SCI-MAP-REQ-033 – MAP-local base-product validity. One MAP-local base-product scientific-validity state shall require membership in the exact support-authorized row set, exact admitted PTC and AST parents, passing MAP contribution/support policy, finite normalized signal, and absence of an applicable MAP-product failure. Explicitly limited or unavailable response/covariance does not alone invalidate the map. Retained storage, finite payload, coefficient, exposure, incidence, normalization, or downstream-local eligibility cannot establish or rewrite MAP validity.

SCI-MAP-REQ-034 – Screen before arithmetic. Invalid contributions shall be excluded before their numerical payload is evaluated. Multiplication by zero, masking after accumulation, or sanitizing a non-finite aggregate shall not substitute for admission.

SCI-MAP-REQ-035 – Exceptional contribution states. Zero coefficient shall be noncontributing, a low positive coefficient shall follow the explicit adopted policy, negative or non-finite ordinary coefficients shall be invalid, and an out-of-bounds projection shall contribute nowhere. Each cause shall remain distinguishable.

SCI-MAP-REQ-036 – Complete observation bundle. A base/unfiltered observation map shall be one immutable ordered bundle containing signal, numerator, normalization identity, response class/state, uncertainty/covariance state, the distinct facts in REQ-025, exposure lineage/accounting identity, unit and exact nominal-beam parent, complete AST frame/WCS, shape/indexing, estimator/projection/profile identities, PTC removed-component/additive-reference/null-space state, lifecycle, exact PTC/AST parentage, and product identity. Limited or unavailable response/covariance states are complete disclosures, not automatic signal invalidity.

SCI-MAP-REQ-037 – Atomic coadd admission. A complete observation bundle shall pass all compatibility and required-companion checks before any coadd-owned state changes. Rejection shall leave numerators, normalizations, counts, exposure, response, validity, provenance, and product cardinality unchanged.

SCI-MAP-REQ-038 – Centered integer common grid. Observation and coadd shapes shall differ by non-negative even row and column counts, with the corresponding integer offset mapping their reference pixels to the same world coordinate. Fractional shift, reprojection, interpolation, wrapping, implicit recentering, and a different sky grid are forbidden.

SCI-MAP-REQ-039 – Ordinary coadd estimator. The exact v0.1 family `SCI-MAP:uniform_observation_coadd_coefficient@1`, owned by MAP, assigns dimensionless $u_{op} = 1$ to every row of each bundle admitted by `SCI-MAP:observation_coadd_admission@1`. It is an equal-observation coefficient, not precision or inverse variance. On exact centered-integer row identities, the coadd accumulates $N_p^c = \sum_o u_{op} \hat{m}_{op}$, $Q_p^c = \sum_o u_{op}$, and publishes $\hat{s}_p = N_p^c / Q_p^c$ only on $\mathcal{S}_{out}^c$.

SCI-MAP-REQ-040 – Coadd companions and unavailable response. The stacked signal vector $\hat{\mathbf{m}}_{obs}$ shall carry explicit observation-row identities; B_{out} is the exact centered-integer equal-observation averaging operator; and T_{obs} is the stacked compatible observation-response object on those same domains. A numerical signal coadd may be produced under the named response-independent base-coadd role when any member response is unavailable, but the coadd response is then typed unavailable and no hidden response subset is formed. Exposure and count use the same atomic admission and placement.

SCI-MAP-REQ-041 – Coadd covariance and precision. C_{obs} shall be the block matrix on exact stacked observation-row identities with every available within- and cross-observation block explicitly located. For fixed admission, $C_{c,out} = B_{out} C_{obs} B_{out}^T$. Missing blocks are unknown, not zero or independence. The uniform observation coefficient is not precision; incomplete covariance limits only claims that require more.

SCI-MAP-REQ-042 – Correlated-GLS boundary. A correlated minimum-variance solution using $K_p^{-1}\mathbf{1}$ may require negative combination coefficients and therefore belongs to a separately authorized correlated-GLS branch, not the ordinary positive-coefficient coadd.

SCI-MAP-REQ-043 – Minimum product identity. Every observation and coadd product shall identify its nonpolarimetric total-intensity-equivalent role; external observation/coadd, array/group, and occurrence domains; estimator and coefficient families; response/covariance states; signal unit and fixed-nominal-beam parent; complete AST frame/WCS; grid/projection/boundary; MAP profiles; exact PTC product/application and coefficient generations; exposure convention; lifecycle; parentage; causes; and product identity. No STOKES token supplies quantity authority. Scientific identity shall be durable without prescribing a persistence format.

SCI-MAP-REQ-044 – Frames and units. MAP shall bind the complete frame identity delivered by `SCI-AST:rtc_output_grid_coordinates@1`: exact source/target frame, reference system (including ICRS or FK5/J2000 as actually declared), equinox/epoch, time scale, site/model state, TAN topology, WCS, axis order, signs, handedness, and degree-valued science axes. MAP shall not equate J2000, FK5, ICRS, apparent, observed, or unrefracted frames by name or default. Signal/response units are `mJy/beam` with exact nominal-beam identity; covariance has its square.

SCI-MAP-REQ-045 – WCS and index persistence. Full-precision typed or sidecar WCS shall be the lossless admission and provenance authority. A physical FITS WCS may differ by at most 0.1 arcsec maximum sky separation while preserving exact axis sign, handedness, orientation, and centered-integer shape/reference-pixel relations. FITS coordinates are one-based and memory indices zero-based.

SCI-MAP-REQ-046 – Realized provenance. Realized state shall record requested, effective, observation-resolved, applied, and realized identities; exact PTC/AST/boundary/profile generations; per-gate outcomes and causes; admitted contributions/bundles; accumulators and row domains; distinct map/exposure facts; PTC λ , $\hat{U}_{\text{corr}}$, $U_{\text{total},\Theta}$, removed-subspace and null-space identities; response class/state-change record; covariance state; offsets; required products; and failures.

SCI-MAP-REQ-047 – Failure scope and atomicity. An invalid required input, incompatible bundle, unrepresentable finite aggregate, unrepresentable coordinate/index conversion, missing required companion, or failed required publication shall propagate at its declared contribution, pixel, observation-admission, or complete-product scope and shall fail before live aggregate state changes.

SCI-MAP-REQ-048 – Required observation products. A complete observation bundle remains required when coaddition is enabled, and every product designated required by the effective v0.1 plan shall be published successfully before completion is asserted. Whether future abstract contracts require physical observation-map publication independent of an effective product policy remains an owner decision.

SCI-MAP-REQ-049 – Immutable base parent. Downstream transformations shall preserve resolvable base/unfiltered MAP-bundle identity and MAP-local base-product validity. A later local validity/eligibility decision may differ for its named use but shall not rewrite MAP validity, alter the original claims, or sever parentage. Later response or covariance estimates are new versioned products.

SCI-MAP-REQ-050 – Complete consumer boundary. No consumer shall receive a bare signal plane as a scientifically complete map. The boundary shall include quantity and nominal-beam identity, estimator/normalization/coefficient/profile identity, honest response class/state and uncertainty/covariance disclosure, distinct map and exposure facts, PTC additive-reference/removed-component/null-space state, exact frame/WCS, shape/indexing, lifecycle, parentage, dependency versions, and failures. Later analyses choose their own claim requirements and may attach honest versioned response/covariance products; MAP does not own their eligibility.

SCI-MAP-REQ-051 – Same fixed noise operator. An admitted noise realization may be described as using the same map/coadd operator only when eligibility, projection, grouping, coefficients, normalization, exact support-authorized row selection, boundary, and atomic coadd admission are fixed or identically prescribed. Re-estimation from the randomized data defines a different named estimator.

SCI-MAP-REQ-052 – Method and claim separation. This authority shall not be applied by analogy to JINC, maximum-likelihood mapmaking, OOF residual transfer, reprojection, filtering, empirical noise, fitting, Beammap inference, feedback, Stokes Q/U, measured-R mapmaking, or other units. Algebraic contract correctness, engineering conformance, representation/response fidelity, observational performance, and production readiness shall remain separate claims.

D Falsifiable predictions

These predictions are preregistered contract consequences or conformance challenges. Their presence is not validation evidence.

SCI-MAP-PRED-001 – Uniform constant. For fixed membership, a constant admitted input with uniform positive coefficients produces that constant on every support-authorized raw-valid output row; numerator and normalization scale together, and no unsupported scientific row is created. *Primary claim layer: Algebraic contract.*

SCI-MAP-PRED-002 – Unequally weighted constant. For fixed membership, unequal positive coefficients still preserve the same constant on every support-authorized output row, while normalization and conditional variance generally change with position; rows outside that set have no normalized scientific value. *Primary claim layer: Algebraic contract.*

SCI-MAP-PRED-003 – One-pixel hand calculation. For an authorized row with admitted z_i and $a_i > 0$, the exact triplet is $(N, Q, \hat{m}) = (\sum a_i z_i, \sum a_i, \sum a_i z_i / \sum a_i)$; an unauthorized row has no normalized scientific value. *Primary claim layer: Algebraic contract.*

SCI-MAP-PRED-004 – One-hot boundary. A finite coordinate inside exactly one half-open cell has one $G_{pi} = 1$, preserves unity over pixels, and contributes only there. A coordinate on an internal upper edge belongs to the adjacent cell; a coordinate on the outer upper extent or outside the grid contributes nowhere. Fractional splitting is not authorized. *Primary claim layer: Algebraic contract.*

SCI-MAP-PRED-005 – Delta input. A basis-pixel sky input produces the corresponding support-row-restricted column of $R_{out} = A_{out}H$, including projection and boundary truncation; it need not be a delta in the output map, and no value is predicted outside the authorized rows. *Primary claim layer: Representation fidelity.*

SCI-MAP-PRED-006 – Point-source template. A declared point-source template τ_b produces $R_{out}\tau_b$ on exactly the support-authorized rows; peak, integrated, and fitted amplitudes are distinct functionals and need separate criteria. *Primary claim layer: Representation fidelity.*

SCI-MAP-PRED-007 – Resolved and Fourier inputs. Resolved disks, gradients, and Fourier modes produce their explicit $R_{out}\tau$ on the authorized row set; a constant-input pass does not predict preservation of these modes. *Primary claim layer: Representation fidelity.*

SCI-MAP-PRED-008 – Variable coverage. Fixed-coefficient constant preservation can coexist with spatially varying normalization, support-row membership, covariance, and nonconstant-template response; a row that fails support is absent rather than measured zero. *Primary claim layer: Algebraic contract.*

SCI-MAP-PRED-009 – Finite boundary. At an admitted finite edge, no contribution wraps or clamps; local normalization may preserve a constant while compact-source peak or integrated response changes. *Primary claim layer: Representation fidelity.*

SCI-MAP-PRED-010 – Coefficient states. Until an exact PTC-owner-selected MAP-facing coefficient family is bound, the ordinary numerical route is unavailable. Under a future exact family, zero is noncontributing, negative/non-finite is invalid, and every state retains its cause; none is inferred as unity or precision. *Primary claim layer: Engineering conformance.*

SCI-MAP-PRED-011 – Use-qualified exclusion and non-finite states. A candidate ineligible under the exact MAP admission or contribution policy for its preserved cause, or failing any required factored gate, does not contaminate an accumulator because exclusion occurs before payload evaluation. No universal flagged-candidate action is inferred. *Primary claim layer: Engineering conformance.*

SCI-MAP-PRED-012 – Threshold selector. For each predeclared positive finite coefficient population, the selected zero-based index is exactly $\lfloor ([0.75N] + N)/2 \rfloor$; empty input yields threshold zero, but no pixel gains support without its own finite positive coefficient. The dimensionless identity of c is fixed; any exact numerical c value not explicitly admitted by the owner-authorized effective policy fails closed before a support row set is constructed, and no numeric domain is inferred. *Primary claim layer: Algebraic contract.*

SCI-MAP-PRED-013 – Missing response. A missing response parent yields a typed unavailable response state, never a fabricated zero or identity response. The original MAP bundle remains usable for claims that do not

require response; any later simulated response or corrected map has a new versioned identity bound to that bundle. *Primary claim layer: Engineering conformance.*

SCI-MAP-PRED-014 – Atomic rejection. A missing required companion or any incompatible identity, unit, response, WCS, shape, policy, or parent rejects the entire observation before every coadd accumulator and count, including apparently compatible pixels, is changed. *Primary claim layer: Engineering conformance.*

SCI-MAP-PRED-015 – One-observation coadd. A coadd containing one admitted observation under the uniform dimensionless coefficient family reproduces its signal on corresponding centered-integer authorized rows. It reproduces response only when the exact response companion is available; otherwise response remains typed unavailable. *Primary claim layer: Algebraic contract.*

SCI-MAP-PRED-016 – Uniform observation coefficients. On each authorized coadd row, two compatible observations produce their equal-observation arithmetic mean and count-like normalization under SCI-MAP: `uniform_observation_coadd_coefficient@1`. The normalization is not precision. *Primary claim layer: Algebraic contract.*

SCI-MAP-PRED-017 – Missing or invalid observation. An unsupported observation row contributes at no coadd row; an incompletely admitted observation contributes nowhere and does not increment admitted-observation count. Neither case creates a zero-valued scientific substitute. *Primary claim layer: Engineering conformance.*

SCI-MAP-PRED-018 – Centered integer placement. Compatible shape differences that are nonnegative and even produce the exact half-difference row/column offset; odd, negative, fractional, or WCS-inconsistent relations fail with no mutation. *Primary claim layer: Engineering conformance.*

SCI-MAP-PRED-019 – Cross-observation covariance. For two maps with variances σ_1^2, σ_2^2 and covariance c , the propagated uniform-coadd covariance includes c . A missing cross block remains unknown, never zero or independence, and the equal-observation normalization is not precision. *Primary claim layer: Algebraic contract.*

SCI-MAP-PRED-020 – Fixed noise operator. A deterministic admitted realization propagated with fixed state yields the same support-row identities and coefficient-by-coefficient A_{out} and B_{out} as the signal path; recomputed state exposes a named difference. *Primary claim layer: Engineering conformance.*

SCI-MAP-PRED-021 – Sequential and parallel reduction. Integer facts, identities, masks, support, and cardinality agree exactly. Floating sums agree with a preregistered operation-count/conditioning bound that is frozen before results are viewed and is not loosened after failure. *Primary claim layer: Engineering conformance.*

SCI-MAP-PRED-022 – No false significance. A deliberate attempt to label signal times square root of merely formal weight as empirical significance is rejected or remains unmistakably labeled formal standardized signal. *Primary claim layer: Engineering conformance.*

SCI-MAP-PRED-023 – No validity promotion. A downstream product may have its own use-local validity or eligibility but cannot change MAP-local base-product validity or sever the base/unfiltered MAP parent. *Primary claim layer: Engineering conformance.*

SCI-MAP-PRED-024 – No ordinary predicate for JINC. A deliberate attempt to apply positive-coefficient ordinary admission, support, or complete-bundle availability to JINC is rejected as outside SCI-MAP v0.1. *Primary claim layer: Engineering conformance.*

SCI-MAP-PRED-025 – Aggregate and index failure. An exact-zero, non-finite, overflowed, or unrepresentable aggregate/index never becomes a live valid product; failure occurs at the declared scope before mutation or completion. *Primary claim layer: Engineering conformance.*

E Product vocabulary

Product	Unit	Contract meaning
Signal numerator	coefficient times mJy/beam	Pre-normalized accumulator; not a sky estimator.
Normalization	declared gridding unit	Positive denominator; precision only under Requirements 020–021.
Normalized signal	mJy/beam	Conditional ordinary map estimand whose vector contains exactly the effective-policy-authorized output rows; no unsupported zero-filled semantic extension.
Formal diagonal weight	inverse square signal unit	Conditional marginal inverse variance under named assumptions; not empirical significance.
Conditional covariance	square signal unit	Full fixed-state noise propagation on the exact scientific output-row domain, with omissions and representation status named.
Response/kernel	mJy/beam when unit-bearing	Realized template response only when the sample-parent identity is proven; otherwise tracer or unavailable.
Geometric/estimator hits	integer	Counts of declared populations, not coverage, precision, or validity.
Exposure facts	named time/accounting unit	Upstream-eligible and retained accounting measures, not precision.
Support facts	logical	Separate normalization and science-policy predicates; their effective conjunction defines output-row membership.
MAP-local base-product validity	logical	Authority for the base/unfiltered MAP bundle only; downstream uses own their eligibility.

F Admitted source register and independence statement

The scientific author used only the following admitted sources:

Source	Exact admitted identity
Scope Brief	SCOPE_BRIEF.md SHA-256 e2a9eb51edb5956191813b4cbbd23866e875d52cdf89cd8b6c272988b4f26674.
Supersession Cover	AUTHOR_SUPERSESSION_COVER.md SHA-256 8ea283525f18199d9760c3f672d145d71f0db87b320ab2e88a5c6635ef3d4aa0.
Frozen core	Exact Git object c28f18ed089657dae278caba2d6d6d65c7ec72f4: doc/audits/packages/SCI-MAP-001_INDEPENDENT_CORE.tex SHA-256 13dd5922bd492e381afcc3b015284216dde1ccc2199ece3d070ee577c7324381.
Conventions and ownership	AUTHOR_CONVENTIONS_AND_OWNERSHIP.md SHA-256 2d478cb6c5e897308d19614b8b01663318744971850c67459f84c7ddcd57c5c9.
Owner handoff correction	CHANGE_LOG_R0.4.md, recording the 2026-08-26 owner-authorized bounded PTC-to-MAP correction.

No implementation, test, validation, reduction, production-status record, MAP-002/MAP-003 evidence, or web source entered this authorship. The bounded handoff review supplied only the owner-authorized authority corrections recorded above. The frozen ordinary derivation was consolidated rather than rederived for novelty.